

Alessandro Metelli

Software Engineer — Backend · Developer Infrastructure · AI Tooling

Remote / GMT-compatible | ametel.dev | github.com/ametel01 | linkedin.com/in/alex-metelli-344169182 | x.com/ametel01

SUMMARY

Software Engineer with 5 years of experience building backend systems, developer infrastructure, blockchain tooling, DeFi protocols, and AI-assisted engineering workflows. Strong in Rust, TypeScript, Python, Go, distributed systems, and developer productivity tooling. Built infrastructure across the Starknet ecosystem and developed full-stack products and protocol tooling spanning validation, automation, and controlled execution. Seeking fully remote backend, platform, developer tools, or AI tooling roles.

EXPERIENCE

Nethermind — Software Engineer

Apr 2024 – Jun 2026 · Remote

- Built backend systems and developer tooling in Rust, TypeScript, Python, Cairo, and Solidity for infrastructure, verification, automation, and developer workflows.
- Developed AggSandbox, a Rust CLI for local multichain infrastructure, transaction execution, bridge testing, and debugging across Starknet, AggLayer, and EVM environments.
- Maintained Voyager Verifier, improving Starknet contract verification through compiler compatibility work, metadata validation, clearer diagnostics, and workflow reliability improvements.
- Contributed `sncast verify` functionality to Starknet Foundry, expanding contract verification support for Starknet developers.
- Built integrations across Starknet, LayerZero, Hyperlane, AggLayer, and EVM ecosystems for cross-chain messaging, local testing, and infrastructure validation.
- Developed ZK infrastructure components, including Groth16 verification and Ethereum-to-Starknet data proving systems.
- Built internal AI-assisted engineering workflows focused on automation, validation, traceability, and developer productivity.

ScopePilot — Founder & Lead Developer

2026 – Present · Remote

- Built an agency proposal scope and pricing configurator with proposal workflows, pricing inputs, client delivery, approvals, revisions, and reporting.
- Implemented a TypeScript/Bun product with Astro, React, Hono, PostgreSQL, Drizzle, Stripe billing, Cloudflare deployment, and automated quality gates.

SELECTED PROJECTS

ScopePilot: Agency proposal scope and pricing configurator with private proposal delivery, workspace branding, client management, billing, reporting, and launch-readiness validation. GitHub. **Tech:** TypeScript, Bun, Astro, React, Hono, PostgreSQL, Drizzle, Stripe, Cloudflare

Horizon Protocol: Pendle-style yield-tokenization protocol for Starknet with PT/YT mechanics, AMM trading, router flows, frontend, indexer, and mainnet deployment support. GitHub. **Tech:** Cairo, TypeScript, React, PostgreSQL, Docker, Starknet

weld-cli: Human-in-the-loop AI coding harness for traceability, validation, controlled execution, and safer AI-assisted development. **Tech:** Python

AggSandbox: Rust CLI for local multichain infrastructure, bridge testing, transaction inspection, and cross-chain debugging. **Tech:** Rust

OPEN SOURCE

- Starknet Foundry: contract verification tooling and developer workflow improvements.
- Voyager Verifier: Starknet smart contract verification infrastructure and diagnostics.
- Apibara: Rust-based blockchain indexing infrastructure and event-processing systems.
- Dojo Engine: tooling and integrations for Starknet application development.

TECHNICAL SKILLS

Languages: Rust, TypeScript, JavaScript, Python, Go, Solidity, Cairo, Noir

Backend: Node.js, REST APIs, PostgreSQL, Redis, event-driven systems, distributed systems

Infrastructure: Docker, Linux, GitHub Actions, CI/CD, AWS **Frontend:** React, TypeScript

AI Tooling: Agent workflows, tool orchestration, browser automation, repository automation, human-in-the-loop systems

Blockchain: Starknet, Ethereum, Aztec, LayerZero, Hyperlane, AggLayer, smart contracts, DeFi, ZK systems

EDUCATION

BSc Data Science and Computing — Birkbeck, University of London